

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

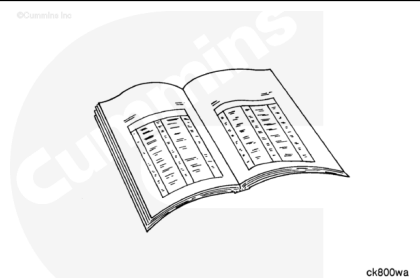
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3
(</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)



with Electronically Actuated Injector

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters, and priming the fuel system so that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations



- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6
(</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3
(</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)

Remove

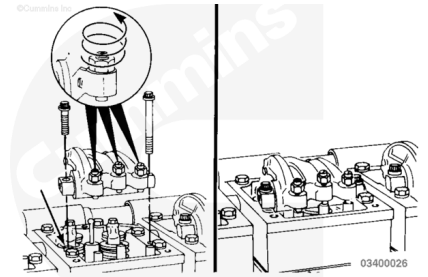
with Mechanically Actuated Injector

Loosen all the adjusting screw nuts and screws to remove all the load from the push rod.

Do **not** allow the rocker levers to fall off the shaft during removal.

Remove the two rocker lever assembly mounting capscrews.

Remove the rocker lever assembly.

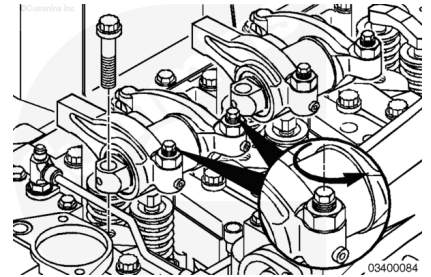


with Electronically Actuated Injector

Loosen the rocker lever adjusting screws.

Loosen the rocker lever shaft mounting capscrews and remove the rocker lever assembly.

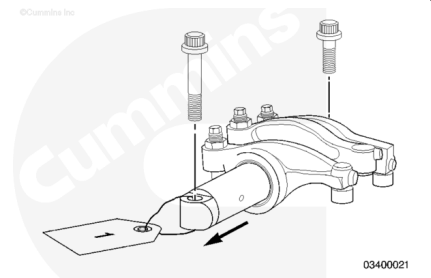
Do **not** allow the rocker levers to fall off the shaft during removal.



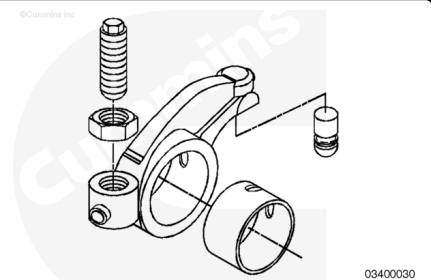
Disassemble

with Mechanically Actuated Injector

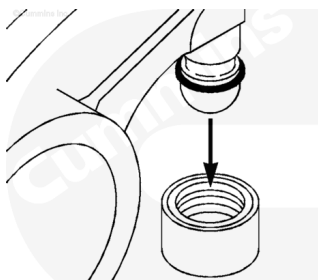
Remove the capscrews. Pull the shaft out. Tag the parts to aid in the assembly.



Remove the locknut and adjusting screw.



Remove the socket from the ball on the rocker lever.



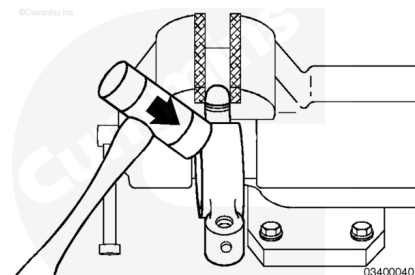
03400039

Note : The ball does not need to be removed unless it or the socket is worn.

Place the ball in a vise as shown.

Wiggle the rocker lever until the ball swivels.

Use a mallet to tap the rocker lever until the ball comes out of the rocker arm.

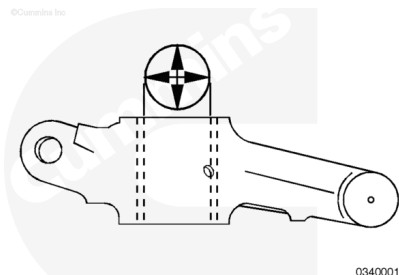


03400040

Check the bushing wear by measuring the inside diameter of the bushing.

Rocker Lever Bushing Bore Inside Diameter (Bushing Installed)

mm		in
47.66	MIN	1.876
47.73	MAX	1.879



03400010

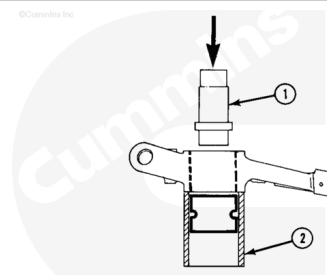
This specification applies to all rocker levers.

Do **not** remove the bushing unless it is damaged or worn.

If cracks are suspected in the rocker lever, remove the bushing to perform magnetic crack inspection.

Support the lever (2) as shown. Use an arbor press and rocker lever mandrel, Part Number 3162458, (1) or equivalent.

Remove the bushing.



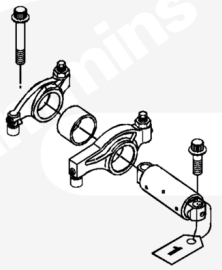
03400047

with Electronically Actuated Injector

Remove the capscrews.

Pull the shaft out of the rocker levers.

Remove the adjusting screws.



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Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

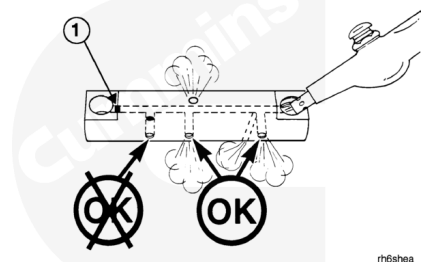
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the shaft.

Use compressed air to blow through the oil drillings, to be sure that the nine oil holes are **not** blocked.

The plug (1) in the main rocker lever shaft **must** be installed in the ring dowel end of the shaft (exhaust lever end).

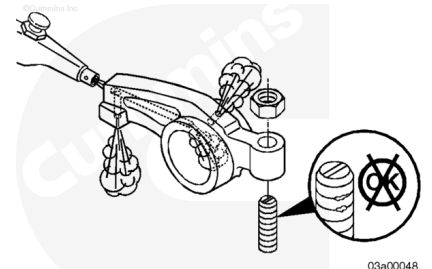
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⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use an abrasive pad to remove any burrs from the bore of the rocker lever.

Clean the rocker lever support with a steam cleaner or solvent.

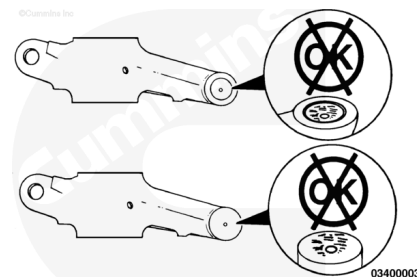
Use compressed air to blow through the oil drillings to be sure the drillings are **not** restricted.

Inspect the adjusting screw threads for damage. Replace if needed.

Inspect the socket for wear on both the injector and valve sockets.

Inspect the crossheads for wear. Refer to Procedure 002-001 in Section 2 (/qs3/pubsys2/xml/en/procedures/20/20-002-001.html)

If there is damage that can be felt with the fingernail, the parts **must** be replaced.



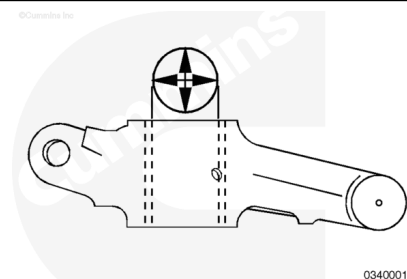
Inspect the bushing for wear.

Measure the inner diameter of the bushing.

Rocker Lever Bushing Bore Inner Diameter (Bushing Installed)

mm		in
47.66	MIN	1.876
47.73	MAX	1.879

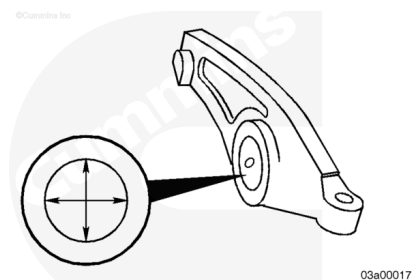
This specification applies to all rocker levers.



Measure each rocker lever bore inside diameter.

Rocker Lever Bore Inside Diameter (Bushing Removed)

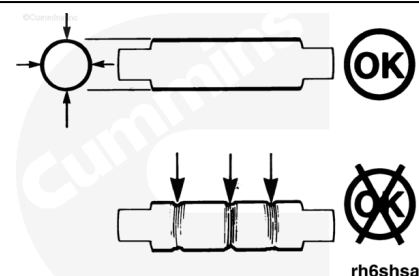
mm		in
49.246	MIN	1.939
49.270	MAX	1.940



Note : The rocker lever shaft for the mechanically actuated injector will show a wear pattern as shown in the graphic.

Inspect the shaft for damage or roughness.

Measure the outside diameter.



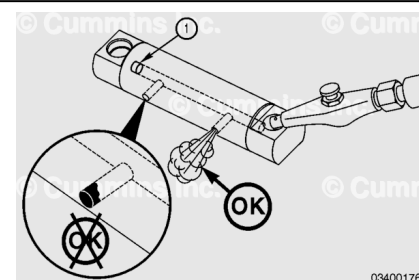
Rocker Arm Shaft Outside Diameter

mm		in
47.592	MIN	1.8737
47.634	MAX	1.8753

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the shaft.

Use compressed air to blow through the oil drillings, to be sure that the nine oil holes are **not** blocked.

The plug (1) in the main rocker lever shaft **must** be installed in the ring dowel end of the shaft (exhaust lever end).

⚠ WARNING ⚠

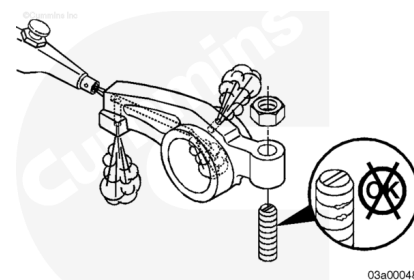
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use an abrasive pad to remove any burrs from the bore of the rocker lever.

Clean the rocker lever support with a steam cleaner or solvent.

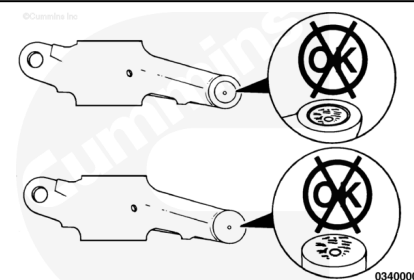
Use compressed air to blow through the oil drillings to be sure the drillings are **not** restricted.

Inspect the adjusting screw threads for damage. Replace if needed.

Inspect the socket for wear on both the injector and valve sockets.

Inspect the crossheads for wear. Refer to Procedure 002-001 in Section 2 (/qs3/pubsys2/xml/en/procedures/20/20-002-001.html)

If there is damage that can be felt with the fingernail, the parts **must** be replaced.

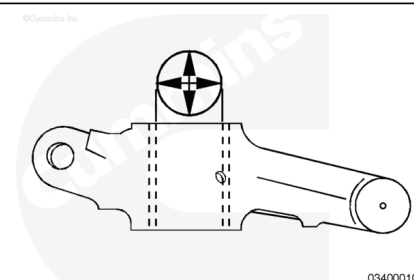


Inspect the bushing for wear.

Measure the inner diameter of the bushing.

Rocker Lever Bushing Bore Inner Diameter (Bushing Installed)

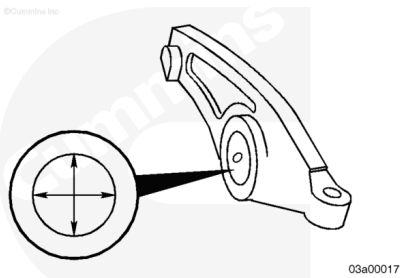
mm		in
34.89	MIN	1.374
34.99	MAX	1.378



This specification applies to all rocker levers.

Measure each rocker lever bore inside diameter.

Rocker Lever Bore Inside Diameter (Bushing Removed)		
mm		in
36.47	MIN	1.436
36.50	MAX	1.437

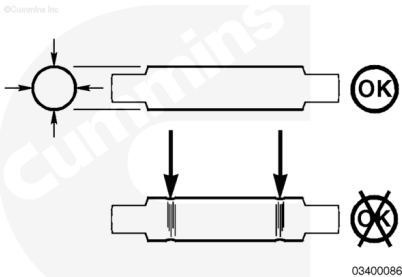


Note : The electronically actuated injector rocker lever shafts will **only** show a wear pattern on the two outer areas of the shaft. See graphic.

Inspect the shaft for damage or roughness.

Measure the outside diameter.

Rocker Arm Shaft Outside Diameter		
mm		in
34.82	MIN	1.371
34.86	MAX	1.373



Magnetic Crack Inspect

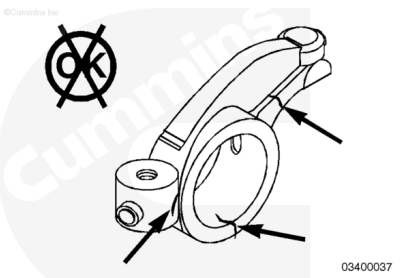
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

The rocker lever **must** be disassembled to perform this inspection.

Use the residual method and apply coil shot amperage.

Ampere turns is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.



Amperage (Ampere Turns)

1200 VDC or rectified VAC

2000 VDC or rectified VAC

The lever **must** be replaced if there is a crack in any location.

Demagnetize the lever.

Use solvent to clean the rocker lever.

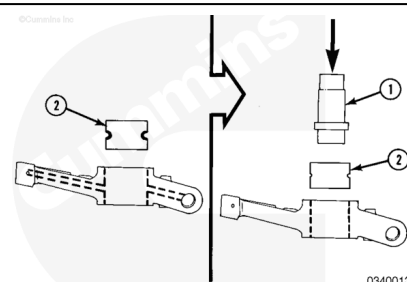
Assemble

⚠ CAUTION ⚠

The bushing must be installed and aligned in the bushing bore correctly. A loss of lubricating oil to the shaft will occur if the bushings are incorrectly installed.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

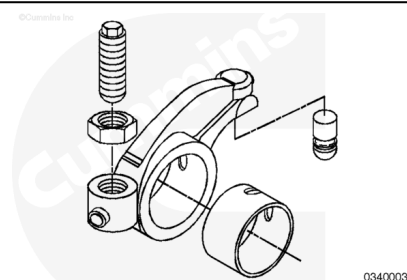


Align the slot in the bushing to point toward the nose of the rocker lever while aligning the oil holes in the bushing with the oil passages in the rocker lever.

Use an arbor press and a rocker lever mandrel (1), Part Number 3162458, for mechanically actuated injectors and Part Number ST-1284, or equivalent for electrically actuated injectors, to install the bushing.

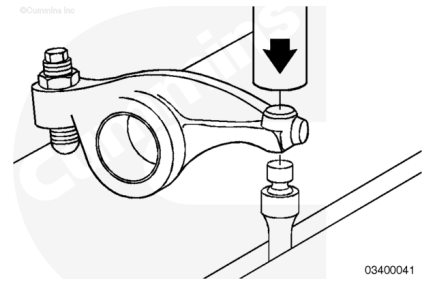
Install the adjusting screw and the locknut in the rocker lever.

Do **not** completely tighten the locknut until the assembly is installed on the engine.



⚠ CAUTION ⚠

Do not overstress the press fit. Damage to the rocker lever can result.



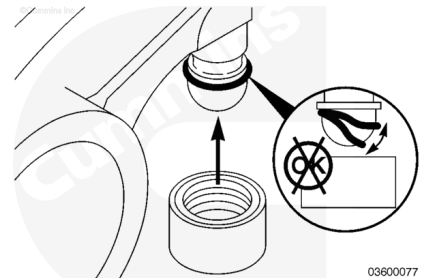
Use an old push tube socket end to support and align the ball for proper seating.

Use a press to seat the new ball into the cavity in the rocker lever.

The lip on the ball **must** be flush with the surface on the rocker lever.

⚠ CAUTION ⚠

Make sure the o-ring is positioned on the ball properly. If the o-ring is installed between the ball and socket the o-ring will eventually fail, resulting in excessive valve lash.

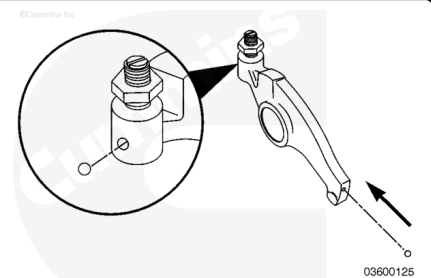


The o-ring is **only** used to retain the socket to the rocker lever during assembly.

Install the new socket on the ball.

If removed, install drive plugs into oil drillings in the rocker levers.

Contact a Cummins® Authorized Repair Location for the proper sealing mechanism part number.

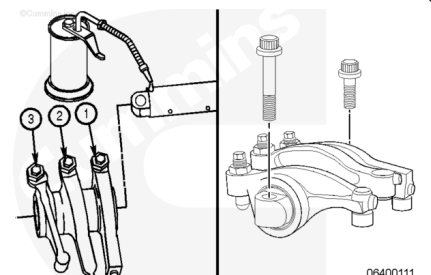


Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.



⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

Use clean oil to lubricate the shaft.

Slide the rocker levers on the shaft in the sequence shown.

- Exhaust lever (1)
- Injector lever (2)
- Intake lever (3)

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

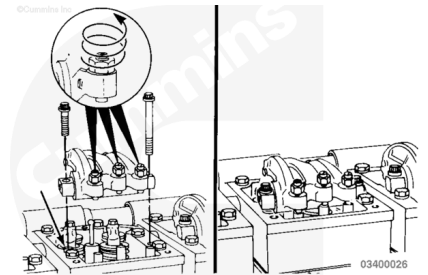
Install the washers and capscrews.

Make sure the ring dowel that aligns the rocker shaft is installed in the housing.

Make sure the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the pushrod sockets with the adjusting screws.



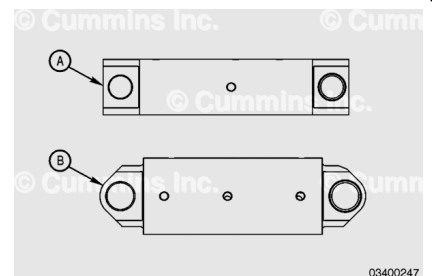
There are two different torque values, depending on the type of rocker lever shaft the engine has.

For capscrews for rocker lever shaft A:

Torque Value: 122 n·m [90 ft-lb]

For capscrews for rocker lever shaft B:

Torque Value: 225 n·m [166 ft-lb]

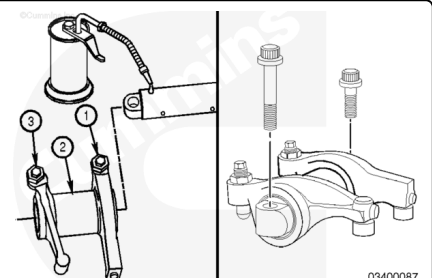


with Electronically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.

⚠ CAUTION ⚠



The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

Use clean oil to lubricate the shaft.

Install the exhaust lever (1), spacer (2), and intake lever (3) onto the shaft. The oil drillings in the shaft **must** align with the drillings in the levers.

Note : The plug in the end of the shaft must be on the same end as the exhaust rocker lever.

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

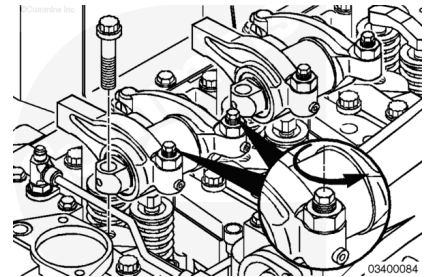
Install the washers and capscrews.

Make sure the ring dowel that aligns the rocker shaft is installed in the housing.

Make sure the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the pushrod sockets with the adjusting screws.



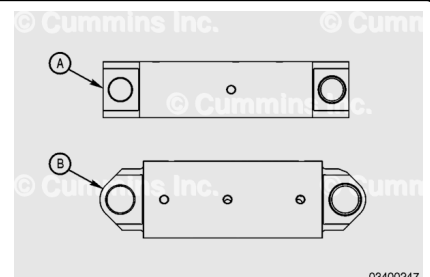
There are two different torque values, depending on the type of rocker lever shaft the engine has.

For capscrews for rocker lever shaft A:

Torque Value: 122 n•m [90 ft-lb]

For capscrews for rocker lever shaft B:

Torque Value: 225 n•m [166 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3
(/qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html)

- Install the rocker lever covers. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Operate the engine and check for leaks.



with Electronically Actuated Injector

- Adjust the overhead. Refer to Procedure 003-006 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Install new injector supply lines. Refer to Procedure 006-051 in Section 6 (/qs3/pubsys2/xml/en/procedures/56/56-006-051.html)
- Operate the engine and check for leaks.

